

Day 15: Lean, Inventory & TQM Master Reference

100-Day Quality Engineering & Industrial Excellence Challenge

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Focus Area:	Core Formulas, Practice Questions with Hints & Lean Concepts	Repository:	github.com/Jagriti1907/100-Day-Quality-Engineering-Challenge-

SECTION 1: STANDARD INDUSTRIAL & COMPETITIVE FORMULA BANK

1. Lean Manufacturing & Line Balancing

Takt Time (T_t)

$$T_t = \text{Total Available Work Time per Shift} / \text{Customer Demand per Shift}$$

Sets the required speed of production to meet customer demand on time.

Cycle Time (T_c) & Minimum Workstations (N_{min})

$$T_c = \text{Available Production Time} / \text{Total Units Made}$$

$$N_{min} = \text{Ceiling} [(\sum t_i) / T_c]$$

t_i is the time needed for each task. Ceiling means rounding up to the next whole number.

Line Efficiency (η) & Balance Delay (BD)

$$\eta = [(\sum t_i) / (N \times T_c)] \times 100\%$$

$$BD = 100\% - \eta = [(N \times T_c - \sum t_i) / (N \times T_c)] \times 100\%$$

Shows how well work is shared across workstations and the percentage of idle time.

Smoothness Index (SI)

$$SI = \sqrt{ \sum (T_c - T_{si})^2 }$$

T_{si} is the total time spent at station i. A lower SI value means a better balanced assembly line.

Overall Equipment Effectiveness (OEE)

$$OEE = \text{Availability (A)} \times \text{Performance (P)} \times \text{Quality (Q)}$$

A = Run Time / Planned Production Time | P = (Ideal Cycle Time × Total Units) / Run Time | Q = Good Units / Total Units

2. Kanban, Pull Systems & Flow Metrics

Kanban Bins Calculation (K)

$$K = [\text{Demand (D)} \times \text{Lead Time (L)} \times (1 + \text{Safety Factor S})] / \text{Bin Size (C)}$$

Calculates how many bins or cards you need to keep production moving smoothly without running out.

Little's Law

$$\text{Manufacturing Lead Time (MLT)} = \text{Work-in-Progress (WIP)} / \text{Production Rate (TH)}$$

Explains how stock on the shop floor directly affects how long orders take to finish.

Process Flow Efficiency

$$\text{Flow Efficiency (\%)} = [\text{Value-Added Time} / \text{Total Lead Time}] \times 100\%$$

Measures the percentage of total time actually spent working on the product.

3. Inventory Planning & Cost Control

Economic Order Quantity (EOQ)

$$EOQ = \sqrt{[(2 \times \text{Annual Demand } D \times \text{Order Cost } S) / \text{Holding Cost } H]}$$

Finds the best order size to balance ordering costs and stock holding costs.

Economic Production Quantity (EPQ)

$$EPQ = \sqrt{[(2 \times D \times S) / (H \times (1 - d/p))]}$$

Used when making items in-house. p = daily production rate, d = daily usage rate.

Reorder Point (ROP) & Safety Stock (SS)

$$ROP = (\text{Daily Demand } d \times \text{Lead Time } L) + \text{Safety Stock (SS)}$$

$$SS = Z \times \sigma_d \times \sqrt{L}$$

Z is the standard normal value for target service level. σ_d is daily demand variation.

4. Process Capability & Quality Metrics

Process Capability Indices (C_p , C_{pk} , C_{pm})

$$C_p = (USL - LSL) / (6\sigma)$$

$$C_{pk} = \text{Minimum of } [(USL - \text{Mean } \mu) / (3\sigma), (\mu - LSL) / (3\sigma)]$$

$$C_{pm} = (USL - LSL) / [6 \times \sqrt{ \sigma^2 + (\mu - \text{Target } T)^2 }]$$

C_p measures potential capability, C_{pk} accounts for off-center process, C_{pm} considers target alignment.

First Pass Yield (FPY) & Rolled First Pass Yield (RFPY)

$$FPY = \text{Good Units Output} / \text{Total Units Input}$$

$$RFPY = FPY_1 \times FPY_2 \times \dots \times FPY_m$$

RFPY gives the chance that a part goes through all m steps without any defect.

SECTION 2: PRACTICAL & COMPETITIVE PROBLEM SET

Category 1: Competitive Exam Style Questions

Q1 • Line Balancing

Assembly Line Performance Evaluation

An assembly line operates with a cycle time $T_c = 10$ minutes across 4 workstations. The station times are $T_{s1} = 9$ min, $T_{s2} = 8$ min, $T_{s3} = 10$ min, and $T_{s4} = 7$ min.

- Find the Line Efficiency (η).
- Find the Balance Delay (BD).
- Calculate the Smoothness Index (SI).

FORMULA HINTS

Total time $\sum t_i = 9+8+10+7 = 34$ min

$$\eta = [\sum t_i / (4 \times 10)] \times 100$$

$$BD = 100\% - \eta$$

$$SI = \sqrt{ [\sum (10 - T_{st})^2] }$$

Q2 • Production Inventory

In-House Batch Production Sizing

A factory makes parts at 100 units/day. Daily usage is 40 units/day. Batch setup cost is ₹1,200 and holding cost is ₹5 per unit/year. Annual demand $D = 10,000$ units (250 workdays).

- Calculate the Economic Production Quantity (EPQ).
- Find Maximum Inventory Level built up in stock.
- Calculate Total Annual Holding & Setup Cost.

FORMULA HINTS

$$EPQ = \sqrt{ [(2 \times D \times S) / (H \times (1 - d/p))] }$$

Here $d = 40$, $p = 100$

$$I_{max} = EPQ \times (1 - d/p)$$

$$TIC = (D/Q) \times S + (I_{max}/2) \times H$$

Q3 • Reorder Planning

Safety Stock & Reorder Point Calculation

Daily demand averages 80 units with a standard deviation of 10 units. Lead time is fixed at 9 days. For a 98% service level, $Z = 2.05$.

- Calculate required Safety Stock (SS).
- Calculate the Reorder Point (ROP).

FORMULA HINTS

$$SS = Z \times \sigma_d \times \sqrt{L}$$

$$SS = 2.05 \times 10 \times \sqrt{9} = 61.5 \text{ units}$$

$$ROP = (d \times L) + SS$$

$$ROP = (80 \times 9) + SS$$

Q4 • Process Quality

Process Capability Index Analysis

A part limit is set to 20.00 ± 0.12 mm. Sampling gives a mean $\mu = 20.04$ mm, standard deviation $\sigma = 0.03$ mm, and target $T = 20.00$ mm.

- Calculate C_p and C_{pk} .
- Calculate the Taguchi Capability Index C_{pm} .
- Check if the process is properly centered.

FORMULA HINTS

$$USL = 20.12, LSL = 19.88$$

$$C_p = (USL - LSL) / (6\sigma)$$

$$C_{pk} = \min [(USL - \mu) / (3\sigma), (\mu - LSL) / (3\sigma)]$$

$$C_{pm} = (USL - LSL) / [6 \times \sqrt{ \sigma^2 + (\mu - T)^2 }]$$

Category 2: Shop-Floor Operations Analytics

Q5 • Machine Efficiency

Press Machine OEE Evaluation

An 8-hour shift (480 mins) includes 30 mins planned break. Total units produced = 3,600. Ideal cycle time = 6 seconds (0.1 min) per unit. 180 units were rejected during quality check.

- Find Availability (A), Performance (P), and Quality (Q).
- Calculate Overall Equipment Effectiveness (OEE).

FORMULA HINTS

$$\text{Run Time} = 480 - 30 = 450 \text{ mins}$$

$$A = 450 / 480 = 93.75\%$$

$$P = (0.1 \times 3600) / 450 = 80\%$$

$$Q = (3600 - 180) / 3600 = 95\%$$

$$OEE = A \times P \times Q$$

Q6 • Material Flow

Kanban Container Sizing

An engine line uses 1,200 pistons daily. Supplier lead time is 0.25 days. A 20% safety factor ($S = 0.20$) is kept. Each bin holds 30 pistons, costing ₹1,500 each.

- Find the required number of Kanban Bins (K).
- Calculate the total money locked in inventory.

FORMULA HINTS

$$K = [D \times L \times (1 + S)] / C$$

$$K = [1200 \times 0.25 \times 1.20] / 30$$

$$\text{Total Pistons} = K \times 30$$

$$\text{Capital} = \text{Total Pistons} \times ₹1,500$$

Q7 • Shop Flow

Lead Time Reduction using Little's Law

A plant holds a Work-in-Progress (WIP) of 800 units with a production rate of 40 units/hour. Management wants to reduce total lead time to 5 hours.

- Calculate current Lead Time.
- Find the target WIP limit needed to achieve 5 hours lead time.

FORMULA HINTS

$$\text{Lead Time} = \text{WIP} / \text{Production Rate}$$

$$\text{Current LT} = 800 / 40 = 20 \text{ hours}$$

$$\text{Target WIP} = \text{Target LT} \times \text{Rate}$$

$$\text{Target WIP} = 5 \times 40 \text{ units}$$

Q8 • Line Yield

Multi-Step Yield & DPMO Calculation

A line has 5 steps with yields: $\text{FPY}_1 = 0.98$, $\text{FPY}_2 = 0.96$, $\text{FPY}_3 = 0.94$, $\text{FPY}_4 = 0.99$, and $\text{FPY}_5 = 0.95$. Each unit has 8 defect opportunities per step.

- Calculate Rolled First Pass Yield (RFPY).
- Find total line DPMO.

FORMULA HINTS

$$\text{RFPY} = \text{FPY}_1 \times \text{FPY}_2 \times \dots \times \text{FPY}_5$$

$$\text{Total Opportunities} = 5 \times 8 = 40$$

$$\text{Defect Rate} = 1 - \text{RFPY}$$

$$\text{DPMO} = (\text{Defect Rate} / 40) \times 1,000,000$$

SECTION 3: ESSENTIAL LEAN GLOSSARY & KEY ACRONYMS

1. Industry Standard Acronyms

ACRONYM	FULL NAME & CLEAR MEANING
TIMWOODS	The 8 Lean Wastes: Transportation, Inventory, Motion, Waiting, Overproduction, Overprocessing, Defects, and Unused Skills.
SMED	Single-Minute Exchange of Die: A method to reduce machine setup time to under 10 minutes.
VSM	Value Stream Mapping: A visual tool to trace material and information flow from start to finished product.
CONWIP	Constant Work-In-Process: A pull system that keeps total shop-floor inventory fixed at a set limit.
DPMO	Defects Per Million Opportunities: A standard quality metric showing defect rate per million chances.
RFPY	Rolled First Pass Yield: The chance that a product passes through all production steps without any defects.

2. Important Lean Operational Terms

TERM	SIMPLE OPERATIONAL MEANING
Muda, Muri, Mura	Muda: Unnecessary waste Muri: Overburdening staff or machines Mura: Uneven production flow.
Heijunka	Production leveling: Balancing product volume and mix evenly over time to avoid big batch spikes.
Poka-Yoke	Mistake-proofing: Simple checks or fixtures that prevent human errors before they can cause defects.
Andon	Visual signal board: Light or display system that workers use to call for quick help when a problem occurs.
Gemba	The real place: The actual factory floor where work happens and value is created.
Jidoka	Smart automation: Building machines that automatically stop working as soon as an issue is detected.
Mizusumashi	Water Spider: A dedicated operator who supplies parts and materials to workstations so builders never stop.

TERM	SIMPLE OPERATIONAL MEANING
Shojinka	Flexible staffing: Adjusting the number of line workers smoothly based on changing customer demand.

AUTHOR DECLARATION & PROJECT NOTE

This **Day 15 Master Reference Document** was independently compiled and created by **Jagriti Singh** for the 100-Day Quality Engineering Challenge.

"Designed as a clean, practical reference guide combining fundamental quality engineering formulas, solved numerical hints, and core Lean manufacturing terms."

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